



UNIVERSITY OF COLOMBO, SRI LANKA



UNIVERSITY OF COLOMBO SCHOOL OF COMPUTING

BACHELOR OF SCIENCE IN COMPUTER SCIENCE

Academic Year 2020/2021 – First Year Examination – Semester II – 2021



SCS 1213 – Probability and Statistics

TWO (2) HOURS

207

To be completed by the candidate

Examination Index No:

Important Instructions to candidates:

1. The medium of instruction and question is **English**.
2. If a page or a part of this question paper is not printed, please inform the supervisor immediately.
3. Note that questions appear on both sides of the paper. If a page is not printed, please inform the supervisor immediately.
4. Write your index number on each and every page of the Question paper.
5. This paper has **04** questions and **15** pages.
6. Answer **ALL** questions. All questions carry equal marks (25 marks).
7. Any electronic device capable of storing and retrieving text including electronic dictionaries and mobile phones are **not allowed**.
8. **Non-Programmable** calculators are **allowed**.

For Examiner's use only

Question No	Marks
1	
2	
3	
4	
Total	

Question 1

(a) A salesman visits 3 houses to sell two products A and B. The probability that on any visit he makes a sale of product A is $\frac{2}{3}$ and the probability of he makes a sale of product B is $\frac{3}{4}$. If the probabilities of selling products A and B are independent, calculate the probability that the salesman will,

- i). sell both products A and B at the visit of the first house? (3 Marks)

- ii). sell one product at the visit of the first house? (3 Marks)

- iii). make no sale of product B in three houses? (3 Marks)

- iv). make at least one product B in three houses? (3 Marks)

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(b) The following table classifies 1000 persons by their sex and their favour for a certain development proposal.

	Male	Female
Favour	250	450
Oppose	170	130

If a person is selected at random from 1000 persons, find the probability that,

- i). the selected person favours the development proposal given that the person is male.

(2 Marks)

- ii). the selected person is a female given that the person opposes the development proposal.

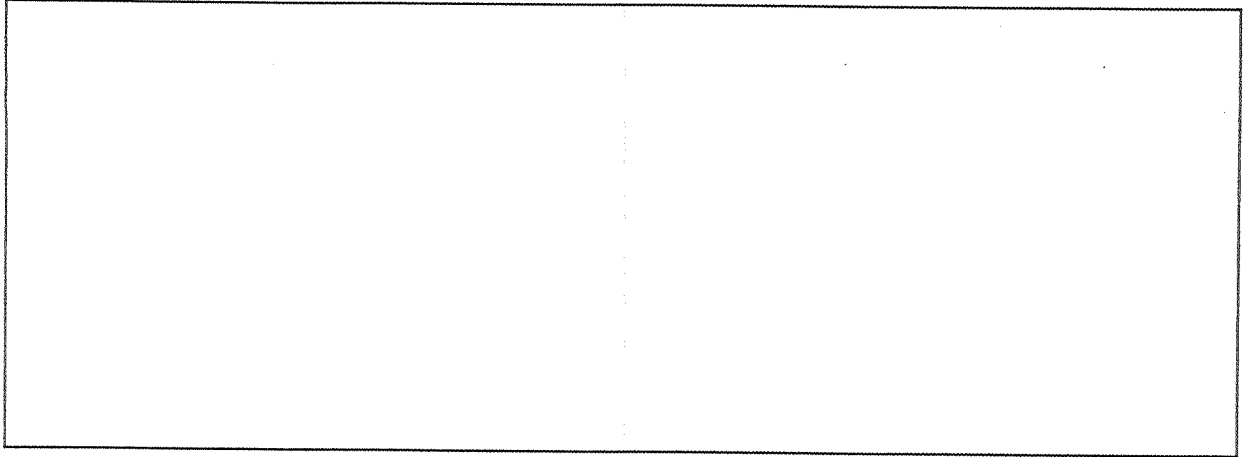
(2 Marks)

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- (c) The probability that a doctor will diagnose a disease X correctly is 0.8. The probability that a patient with disease X will die by his treatment after correct diagnosis is 0.3. The probability that a patient with disease X will die after not diagnosing the disease correctly is 0.7.

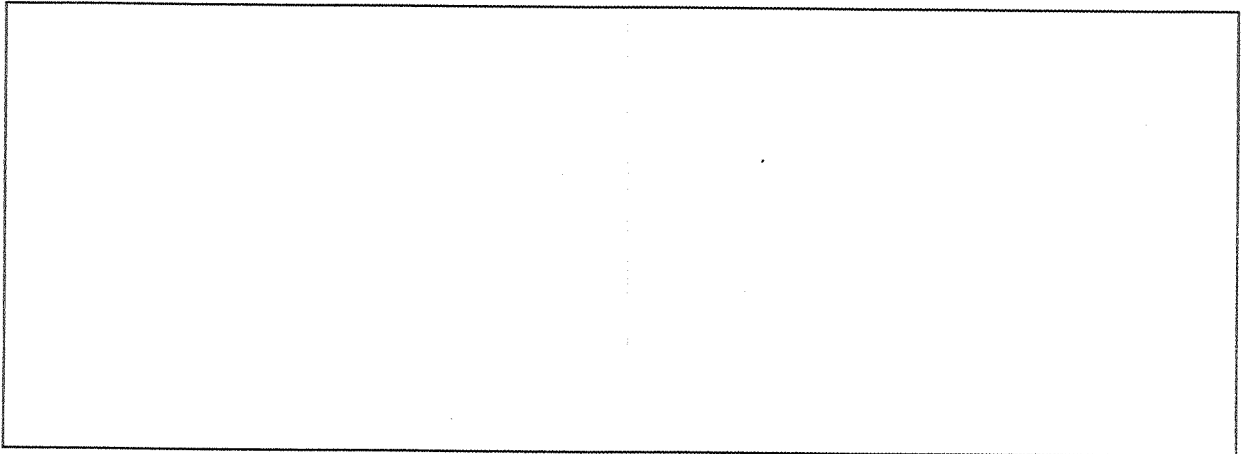
i). What is the probability that a patient will die?

(4 Marks)



ii). If a patient with disease X died, find the probability that the doctor had diagnosed disease X correctly.

(5 Marks)



Question 2

(a) A discrete random variable X has the probability mass function shown in the table below.

X	-1	0	1	2	3
$P(X=x)$	0.25	0.125	k	0.3	0.125

i). Find the value of k .

(2 Marks)

ii). Find the value of $E[X]$.

(2 Marks)

iii). Find the value of $V[X]$.

(3 Marks)

iv). Find the value of $V[2X-3]$.

(2 Marks)

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(b) A website receives hits at a rate of 300 per hour.

- i). State a distribution that is suitable to model the number of hits obtained during a 1-minute interval. (2 Marks)

- ii). State two reasons for your answer to part (a). (2 Marks)

- iii). Find the probability of receiving 10 hits in a given minute. (2 Marks)

- iv). Find the probability of receiving at least 5 hits in 2 minutes. (4 Marks)

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- v). The website will go down if there are more than 70 hits in 10 minutes. Using a suitable approximation, find the probability that the website will go down in a particular 10-minute interval. (6 Marks)

Question 3

(a) An urn contains 3 red balls and 2 blue balls. A ball is drawn at random. If the ball is red, it is kept out of the urn and a second ball is drawn at random from the urn. If the ball is blue, then it is put back in the urn and a red ball is added to the urn, then a second ball is drawn at random from the urn. If X is the random variable 'the number of blue balls drawn',

- i). Find the probability distribution of X by showing the calculations. (5 Marks)

- ii). Find the expected number of blue balls. (2 Marks)

- iii). Find the standard deviation of X . (3 Marks)

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(b) A test consists of 18 multiple-choice questions. For each question, there are four possible answers with only one correct answer. Kamal did not study well for this test. So, he decides to choose the answer to each question randomly. Using an appropriate distribution, answer the following questions.

i). What is the probability that he gets no correct answers?

(2 Marks)

ii). What is the probability that he gets three correct answers?

(1 Mark)

iii). What is the most likely number of correct answers that he gets right?

(3 Marks)

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- iv). Students who get more than half the answers correct, pass this test. What is the probability that Kamal passes? (3 Marks)

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- v). After the test Kamal reckoned that, if he had done 'a little revision', there were four questions which he would definitely have got correct answers and that he would have been able to eliminate one of the wrong answers in each of the other 14 questions, before randomly guessing his answer. Show that with 'a little revision' his chances of passing would have been increased by a factor of more than 50. (4 Marks)

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- vi). Justify the suitability of the selected distribution to calculate the above probabilities. (2 Marks)

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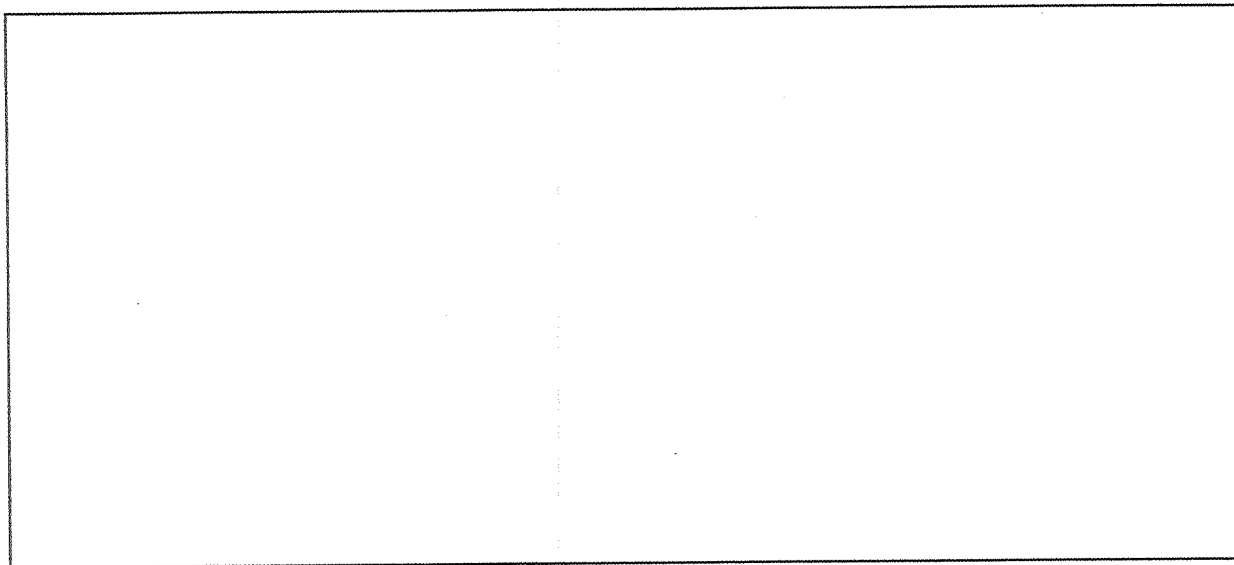
Question 4

(a) The continuous random variable X has probability density function given by

$$f(x) = \begin{cases} 30x^2(1-x)^2 & 0 \leq x \leq 1 \\ 0 & \text{Otherwise} \end{cases}$$

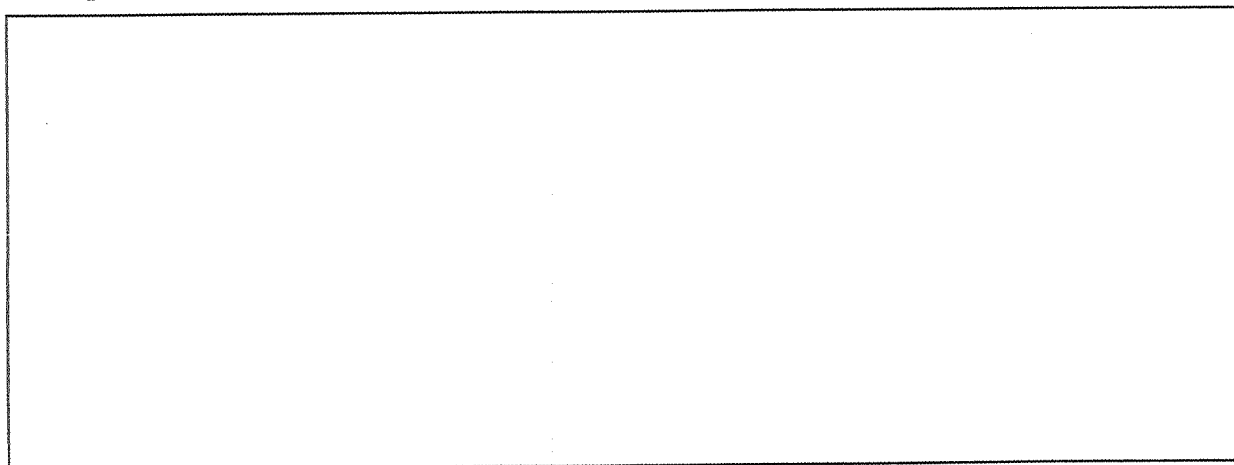
i). Find $E(X)$ and $Var(X)$.

(2 Marks)



ii). A random sample $(X_1, X_2, X_3, X_4, X_5)$ of size 5 is taken from this distribution. Find, correct to 4 decimal places, the probability that all 5 observations exceed $1/3$. Note that, all the 5 observations are independent.

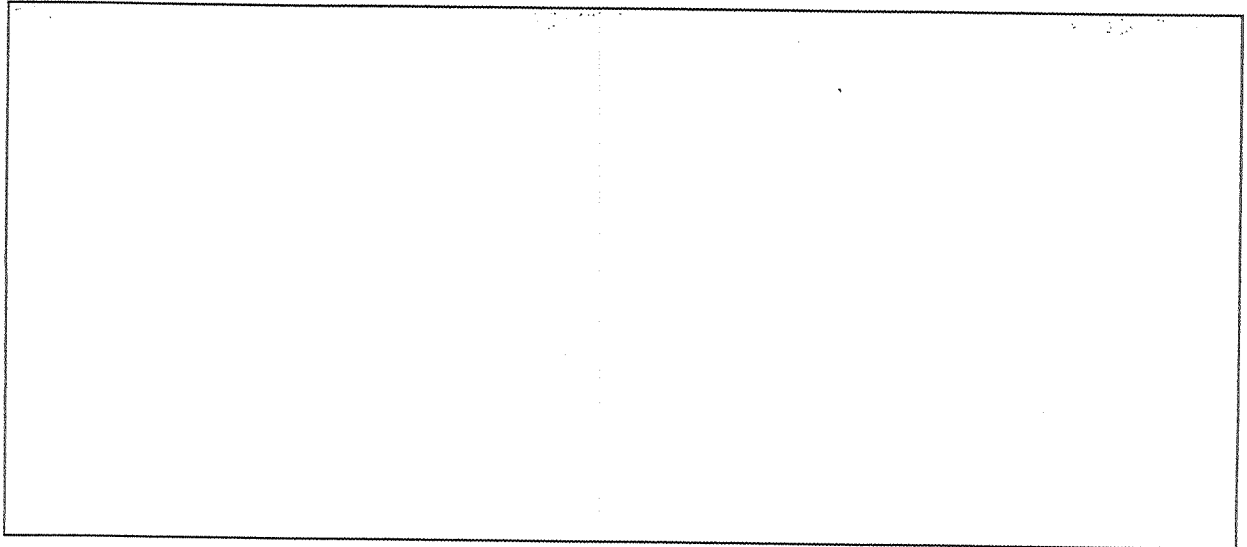
(3 Marks)



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iii). Find the expected value of $g(X) = e^{3X/4}$.

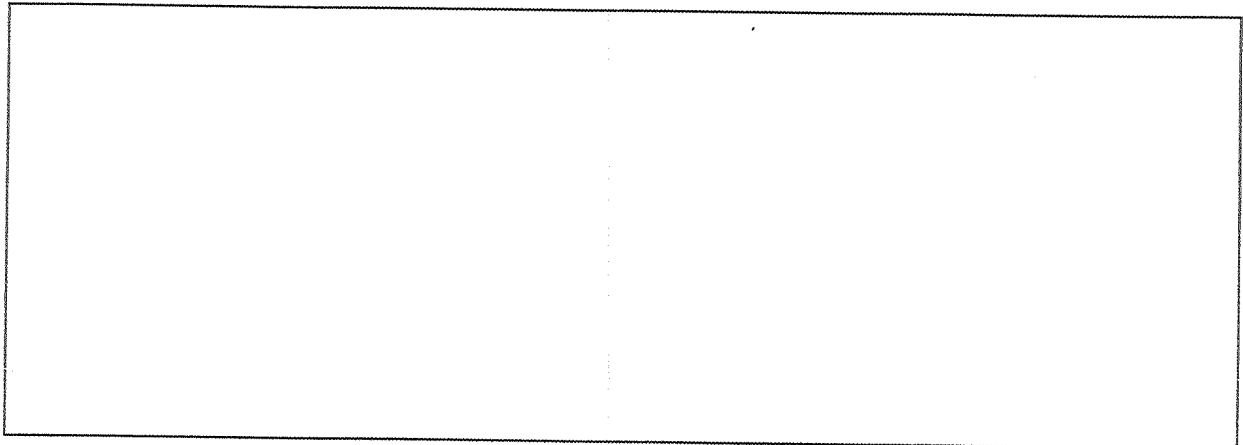
(3 Marks)



(b) An airline operates a baggage weight allowance of 25 kg per passenger. Check-in records show that the actual weight, W kg, of a randomly chosen passenger's baggage can reasonably be assumed to be normally distributed with mean 24 and variance 1.

i). Find $P(W > 25)$.

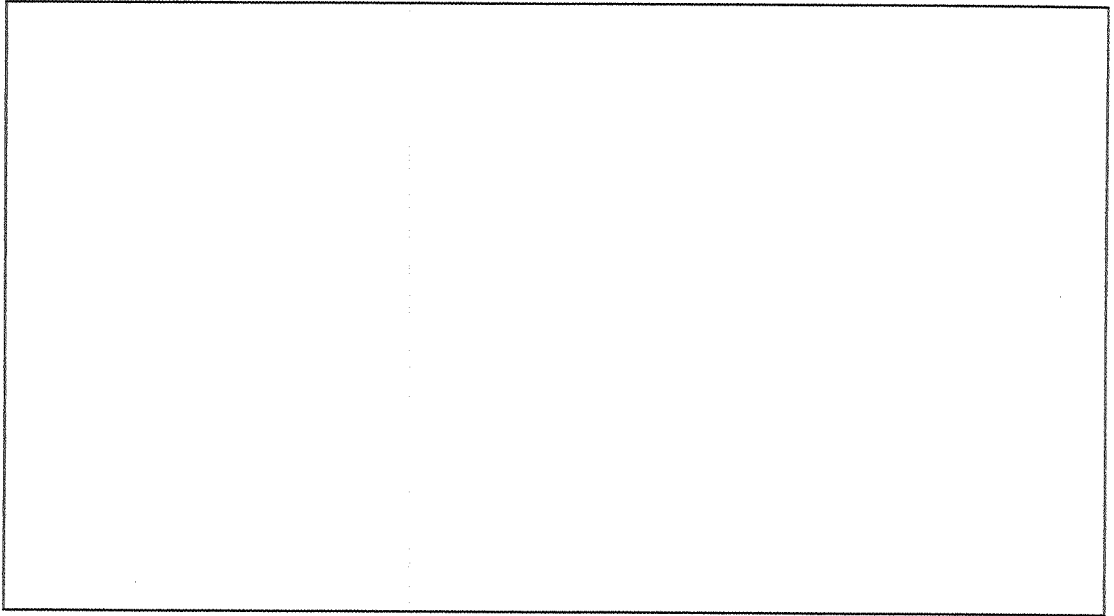
(2 Marks)



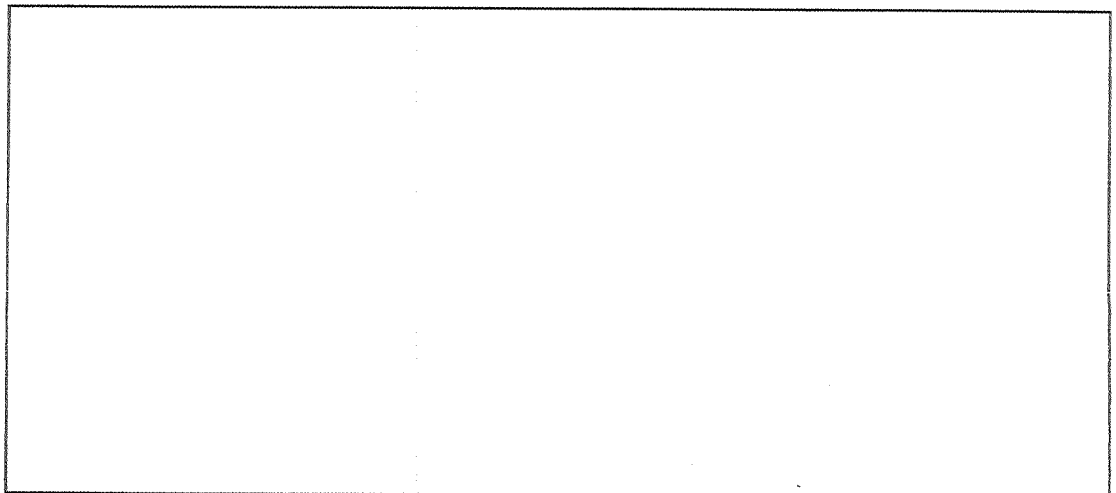
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ii). A passenger with baggage weighing more than 25 kg is charged Rs. 5000 for each kg by which the weight of his or her baggage exceeds 25 kg, all fractions of a kg being rounded up to the next whole number.

- A. If C denotes the excess baggage charge in Rs. 1000 for a randomly chosen passenger, find the probabilities $P(C=0)$, $P(C=5)$ and $P(C=10)$. (3 Marks)

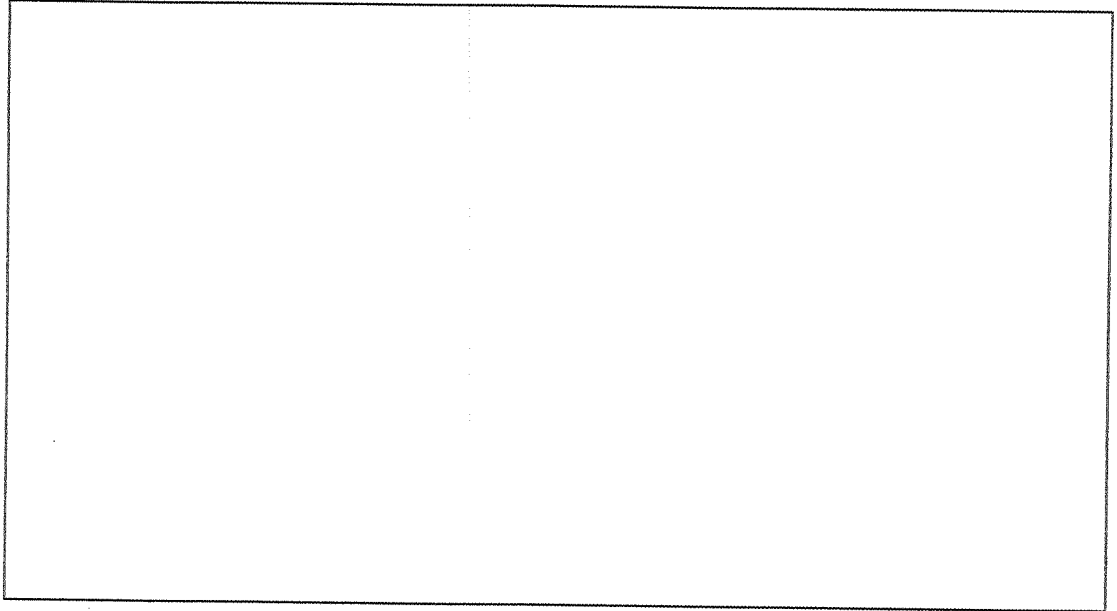


- B. Given that $P(C = 15) = 0.0013$ approximately and that $P(C > 15)$ is negligible, find $E(C)$ and $\text{Var}(C)$. (3 Marks)

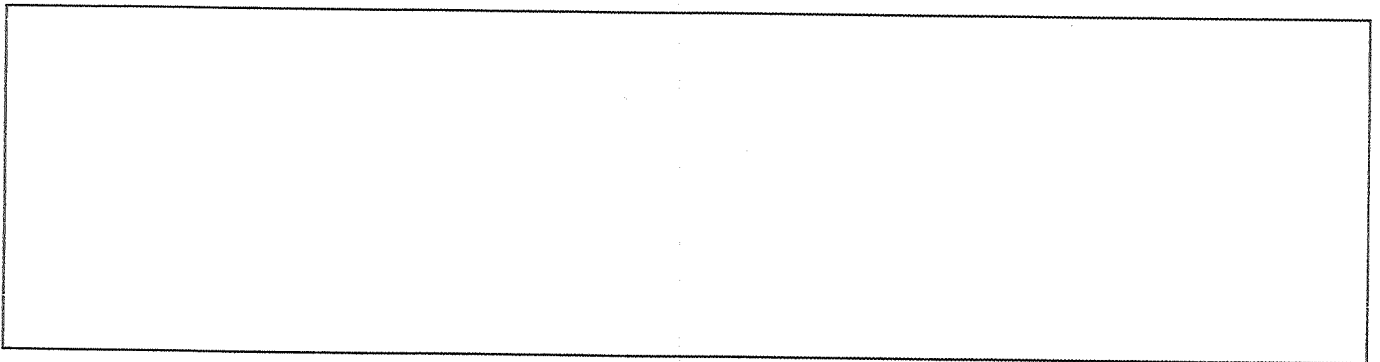


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- C. Assuming that 100,000 passengers independently fly with this airline in a year, write down the mean and variance of C_T , the total excess baggage costs paid to the airline in a year. Use a normal approximation to find the value of C_T which is exceeded with probability 0.05. (5 Marks)

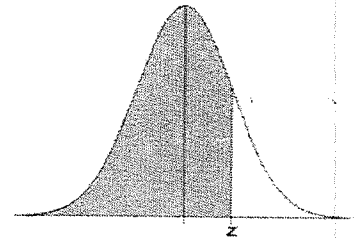


- (c) Eggs are packed into boxes of 500. On average, 0.7% of the eggs are found to be broken when the eggs are unpacked. Using a Poisson approximation, find, correct to two significant figures, the probability that in a box of 500 eggs, at least two are broken. (4 Marks)



Standard Normal Cumulative Probability Table

Cumulative probabilities for POSITIVE z-values are shown in the following table:



z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.9920	0.9922	0.9925	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
2.6	0.9953	0.9955	0.9956	0.9957	0.9959	0.9960	0.9961	0.9962	0.9963	0.9964
2.7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974
2.8	0.9974	0.9975	0.9976	0.9977	0.9977	0.9978	0.9979	0.9979	0.9980	0.9981
2.9	0.9981	0.9982	0.9982	0.9983	0.9984	0.9984	0.9985	0.9985	0.9986	0.9986
3.0	0.9987	0.9987	0.9987	0.9988	0.9988	0.9989	0.9989	0.9989	0.9990	0.9990
3.1	0.9990	0.9991	0.9991	0.9991	0.9992	0.9992	0.9992	0.9992	0.9993	0.9993
3.2	0.9993	0.9993	0.9994	0.9994	0.9994	0.9994	0.9994	0.9995	0.9995	0.9995
3.3	0.9995	0.9995	0.9995	0.9996	0.9996	0.9996	0.9996	0.9996	0.9996	0.9997
3.4	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9998

